FINAL PROJECT REPORT

Solar Spill

Coordinating EV charging on a shared suburban transformer, using live Queensland grid data

April to October 2026

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Repository: github.com/SSV227/solar-spill

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1. Executive summary

Solar Spill began as an EV charging scheduler for Queensland and became a study of a narrower, harder question: **what happens to a shared suburban transformer when every household's EV responds to the same cheap price signal at the same time?**

The project reads live 5-minute Queensland spot price and demand from AEMO, turns each reading into a charge or wait decision, plans each car's charging around a deadline, and simulates 100 EV homes on one 500 kVA distribution transformer under six charging strategies, replaying 352 intervals of real captured grid data. A physical ESP32 demonstrator shows both the live decision and a ten-house version of the main result on a breadboard.

> **Headline result.** Every car independently charging in its own cheapest hours was the **worst** strategy tested: all 100 cars switched on in the same five-minute interval (coincidence factor 1.00), peaking at **157% of the transformer rating with 42 overloaded intervals**, worse than a fixed 11pm timer. Capacity-aware coordination held the peak to **99.9% with zero overloads**, every car still met its deadline, and it cost about **17 cents more per car per night**.

Key findings

- **Individually optimal is collectively worst.** Cars that all read the same price feed all compute the same cheapest hours, so smart charging sharpens the peak instead of spreading it.
- **Statewide signals cannot protect a local transformer.** Randomised staggering on a statewide signal cut the peak to 114% but still left 11 overloads. Only strategies aware of the local transformer cleared it.
- **Smart timing barely saves the driver money.** About 26 cents per car per night on the wholesale component, and nothing on a flat retail tariff. The value of coordination is in the network, not the household bill.
- **Home batteries do not fix the spike.** With half the homes carrying a 10 kWh battery, the overnight EV peak was unchanged.
- **The hardware reproduces the result.** Ten houses replaying a real night: uncoordinated, all 10 charge at once with 42 overloaded intervals; coordinated, never more than 6, zero overloads, all 10 deadlines met.

Status. All five planned phases are complete. The code, results and documentation are public at github.com/SSV227/solar-spill.

2. The problem

Queensland's grid is under pressure at both ends of the day. In the evening, EV owners arrive home and plug in around 5 to 7pm, just as rooftop solar drops away and household demand peaks. In the middle of the day the opposite happens: so much rooftop solar floods the grid that wholesale prices regularly fall to zero or below.

The obvious response is smart charging: shift each car into the cheapest hours. The difficulty is that a price signal is shared. When thousands of cars read the same price feed, they reach the same conclusion at the same moment. Energex and Ergon's *EV SmartCharge Queensland* trial observed exactly this: when one retailer's cheap rate began at a fixed hour, roughly 13% of participants started charging at that exact moment, creating a sharp artificial peak.

That peak lands on the distribution transformer serving a street or suburb. Energex designs its low-voltage network for roughly 3 to 4 kW per house at peak; a single home EV charger draws around 7 kW. Cheap power statewide does not mean one street's transformer can physically deliver it to every house at once.

> **The core distinction this project defends:** a price signal coordinates behaviour. It does not create physical headroom. Wholesale price and local transformer capacity are different constraints, and optimising for the first can break the second.

Research question

Given one shared transformer and a set of EVs that each need a full charge by morning, how do different charging strategies compare on peak load, overloads, cost and deadline compliance, and which strategies, if any, keep the transformer within its rating while every car still finishes on time?

3. Industry context

The market moved substantially during the project, and in a direction that sharpened its framing.

Individual smart charging is now a commodity. Retailer time-of-use tariffs, Amber Electric's price-following and every EV's built-in scheduler already shift a single car into cheap hours. Charge HQ, Australia's best-known standalone smart charging app, was acquired by Amber in early 2025 and closed on 7 September 2026, with Amber citing a shift in focus to vehicle-to-grid and whole-of-site optimisation. Amber's replacement service schedules charging against 5-minute wholesale prices using CSIRO-developed forecasts. A standalone app that times one car against the spot price no longer has a market.

Networks already manage EV chargers. Under the Queensland Electricity Connection Manual (QECM) version 4, effective February 2024, Energex and Ergon offer dynamic EVSE connections: a two-way link between the network and the charger that sends dynamic operating envelopes (DOEs) limiting charging when the local network is stressed. Energex's own modelling suggests active EVSE management could prevent around 75% of zone-substation constraints by 2040. In Queensland, home chargers drawing more than 20 A now require active device management.

Low-voltage visibility is a funded priority. The 2026-27 federal budget allocated $97.2 million to the National Consumer Energy Resources Roadmap. Vendors such as Itron (LV DERMS), EA Technology and Zepben sell distribution networks tools to forecast transformer constraints from coincident EV charging. RACE for 2030 has flagged the fairness of how scarce capacity is shared between customers as an open research direction.

Implication for this project. The value of Solar Spill is not a consumer product. It is a clear, quantified demonstration of the problem that networks and LV-management vendors are now paid to solve, built from real Queensland data, with an honest comparison of strategies.

4. Project timeline

The project ran from April to October 2026 at roughly 2 to 3 hours per week alongside full-time study.

When	Phase	Work completed
April to May 2026	Concept	Problem definition, background research into the Queensland grid, AEMO data sources and the EV SmartCharge Queensland findings. Initial concept and plan.
June 2026	1. Data pipeline	Live AEMO fetch, 5-minute logger, price and demand graphs. First dataset captured (Saturday 27 June). Repository created. Strategic reframe from consumer savings to network coordination.
July 2026	2. Decision engine	Charge/wait logic, Grid Health Score, DECISIONS.md, live scheduler, deadline-aware planner. Weekday dataset captured (2 to 4 July, 352 readings).
July 2026	3. Simulation	100-home, 500 kVA transformer simulation; six strategies; battery scenario; plots and PHASE3.md.
Late July to August 2026	4a. Hardware, live	ESP32-S3 toolchain, Wi-Fi, Flask endpoint, JSON parsing; a single LED driven by live AEMO data. Ten LEDs wired and verified.
September 2026	Review	Market re-analysis after the Charge HQ closure; repositioning around network coordination and industry terminology.
October 2026	4b and 5. Demonstrator and write-up	Ten-house replay demonstrator with smart and coordinated modes; results-first README; repository made public; this report.

5. System architecture

The system is a set of small Python modules, each with one job, feeding two consumers: the offline simulation and the live hardware demonstrator. The ESP32 is deliberately a thin display; all decision and coordination logic stays in Python, where it can be tested without reflashing hardware.

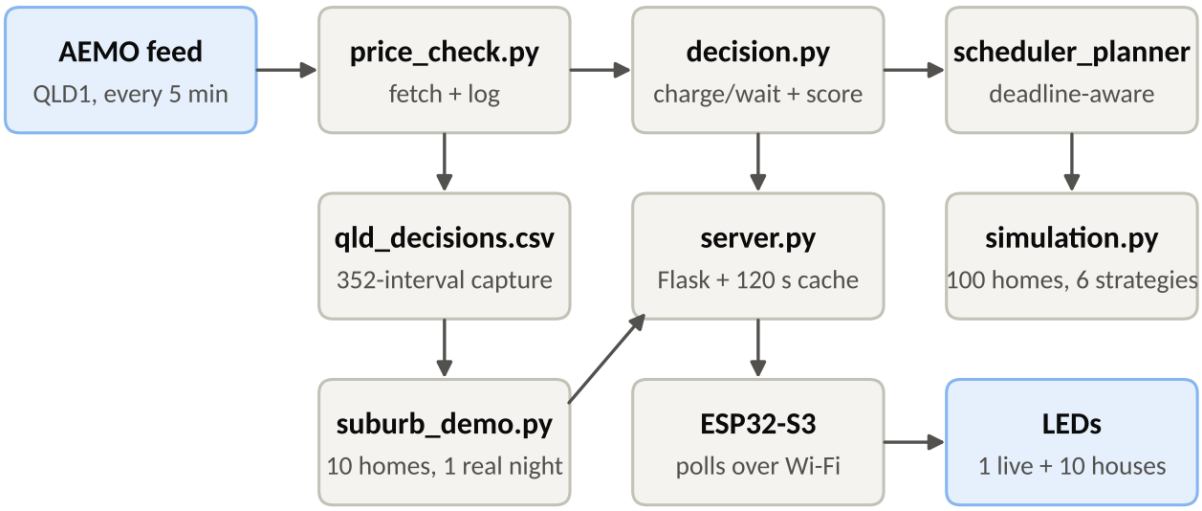


Figure 1. Data flow from the AEMO feed to the simulation and the hardware demonstrator.

Component	Role
price_check.py	Fetches the AEMO ELEC_NEM_SUMMARY feed and extracts the QLD1 spot price, demand and settlement timestamp.
logger.py, graph.py	Polls every 5 minutes, logs to CSV with dropout handling, and plots price against demand.
decision.py	Turns one reading into a CHARGE or WAIT call and a 1 to 10 Grid Health Score.
scheduler.py	Runs the decision engine on a continuous 5-minute loop and logs the results.
scheduler_planner.py	Deadline-aware planner: chooses the cheapest intervals that still deliver a car's required energy before its deadline.
simulation.py	100 homes on one 500 kVA transformer; six strategies; peak, coincidence, overloads, cost, deadlines.
server.py	Flask server: live decision at /decision; ten-house demonstrator at /suburb.
suburb_demo.py	Ten houses replaying one real night under smart and coordinated strategies.
esp32/ sketches	Arduino C++ for the ESP32-S3: Wi-Fi, HTTP polling, JSON parsing, LED control.

6. Phase 1: Live grid data pipeline

AEMO publishes a public JSON summary of the National Electricity Market (ELEC_NEM_SUMMARY), updated every five minutes. The pipeline filters to the Queensland region (QLD1) and keys every reading on the settlement timestamp rather than the local clock, so a delayed poll is still attributed to the correct dispatch interval.

Datasets captured

Dataset	Coverage	What it showed
Saturday capture	27 June 2026	A $32/MWh midday solar low rising to a $130/MWh evening peak; demand roughly 6,800 to 7,474 MW.
Weekday capture	2 Jul 21:00 to 4 Jul 11:35 (352 readings)	Statewide demand peaked near 7,520 MW, well below the 8,000 MW initially assumed. Negative prices on multiple days, down to -$2.50/MWh. The weekday capture drives the simulation and both demonstrators.

What the data changed

- **The cheap window is midday, not overnight.** Solar oversupply produced the lowest and negative prices around the middle of the day. A car on a default overnight timer misses this entirely, which undermines fixed schedules as a strategy.
- **The demand band was recalibrated.** Demand never approached the 8,000 MW ceiling first assumed, so the demand scoring band was revisited.

7. Phase 2: Decision engine and planner

Charge or wait

The charge/wait call is driven by price. A negative price is always CHARGE (the grid is paying to take power). Below $50/MWh is CHARGE. At or above $50/MWh is WAIT. A high-demand warning rides alongside the decision without flipping it on its own.

Grid Health Score

A single threshold is a blunt signal: everything is either on or off. The Grid Health Score gives a graded 1 to 10 measure that later phases use as a probability dial for staggering. It blends two sub-scores, each between 0 and 1:

Sub-score	Definition	Weight
Price	1.0 if negative; 0.0 at or above $100/MWh; linear in between	60%
Demand	1.0 at or below 6,500 MW; 0.0 at or above 8,000 MW; linear in between	40%
Score	round(1 + 9 x blend), clamped to 1 to 10	

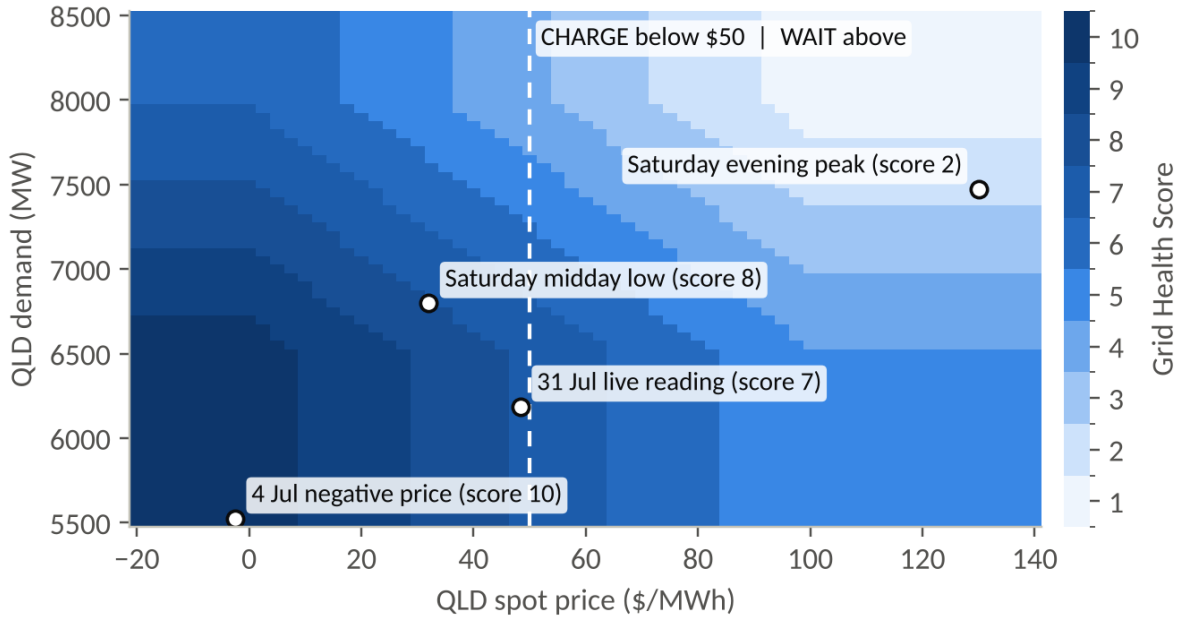


Figure 2. The Grid Health Score across price and demand, computed directly from decision.py, with real readings marked. Darker is a better time to charge.

Price is weighted more heavily because it is the sharper signal of whether to charge; demand moves more slowly and acts as a stress warning. Every threshold and the reasoning behind it is recorded in DECISIONS.md.

Deadline-aware planner

The driver sets only a deadline and a target, for example full by 7am. The planner ranks every 5-minute interval between plug-in and deadline by price and selects the cheapest ones needed to deliver the required energy at the charger's rate. There is no fixed price cut-off: if the best available hours are moderately priced, it uses them, because that is the best it can do while still guaranteeing a full car. This planner is the per-vehicle engine every simulated car runs in Phase 3.

A dead end worth recording

Both obvious rules recreate the spike. A fixed "charge at 11pm" rule moves the evening peak to 11pm. A fixed "charge below $50/MWh" rule switches every charger on the instant price crosses the line. Any single shared threshold makes chargers flip together, which is the problem being solved. This observation is why the design separates per-car planning from coordination across cars.

8. Phase 3: Transformer simulation

Setup

- 100 EV homes on one shared 500 kVA distribution transformer.
- Replays the 352-interval weekday capture of real Queensland prices.
- Each car runs the Phase 2 deadline-aware planner. Chargers draw about 7.4 kW (derived from the simulation output: 740 kW of EV load at a coincidence factor of 1.00 across 100 cars), on top of around 46 kW of household base load at the peak.
- Metrics per strategy: peak transformer load, coincidence factor (share of cars charging in the worst interval), overloaded 5-minute intervals, average wholesale cost per car per night, and the share of cars that met their deadline.

The six strategies

Strategy	Behaviour
Charge on plug-in	Charge immediately on arrival. The baseline.
Fixed 11pm timer	Every car starts at 11pm, like a default vehicle or tariff timer.
Smart, uncoordinated	Each car independently picks its own cheapest intervals before its deadline.
Smart + stochastic stagger	Smart planning, with the Grid Health Score used as a probability dial so cars randomly defer when the grid tightens.
Smart + local-load awareness	Each car sees how loaded its own transformer is and defers when it is near full, overriding only when its deadline forces it. No central authority.
Smart + capacity-aware allocation	A coordinator hands out charging slots cheapest-first but never beyond what the transformer can carry, scheduling the least flexible cars first.

Results

Strategy	Total peak (kW)	% of rating	Overloads	Coinc. factor	Cost / car	Deadlines
Charge on plug-in	755.4	151.1%	38	0.96	$1.86	100%
Fixed 11pm timer	777.6	155.5%	45	0.99	$1.79	100%
Smart, uncoordinated	786.1	157.2%	42	1.00	$1.60	100%
Smart + stochastic stagger	570.4	114.1%	11	0.71	$1.86	100%
Smart + local-load awareness	504.0	100.8%	2	0.62	$1.87	100%
Smart + capacity-aware	499.6	99.9%	0	0.61	$1.78	100%

Source: phase3_out/summary.csv. Cost is the wholesale energy component only.

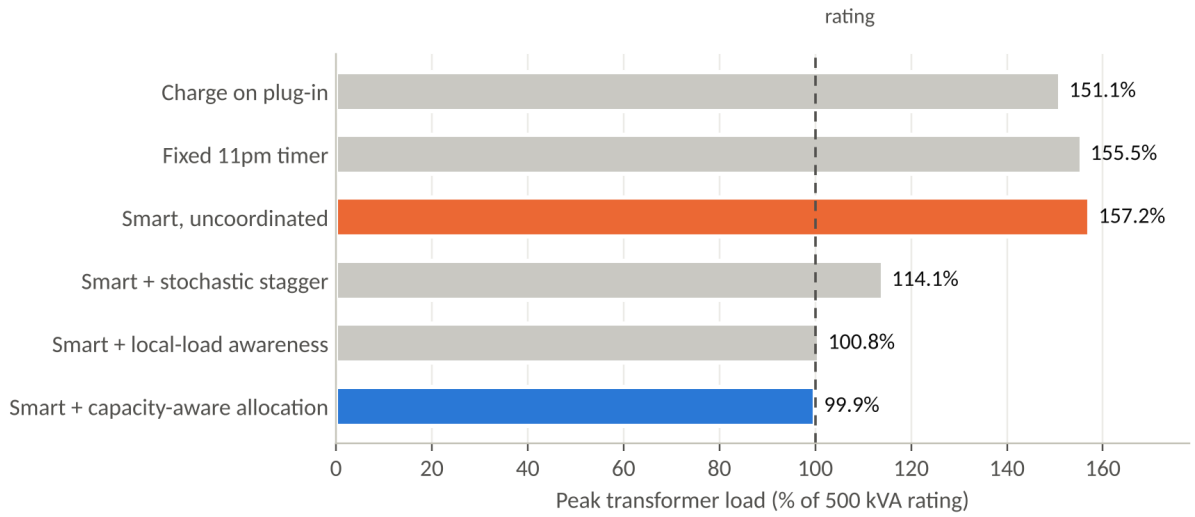


Figure 3. Peak transformer load by strategy. The dashed line is the transformer's rating.

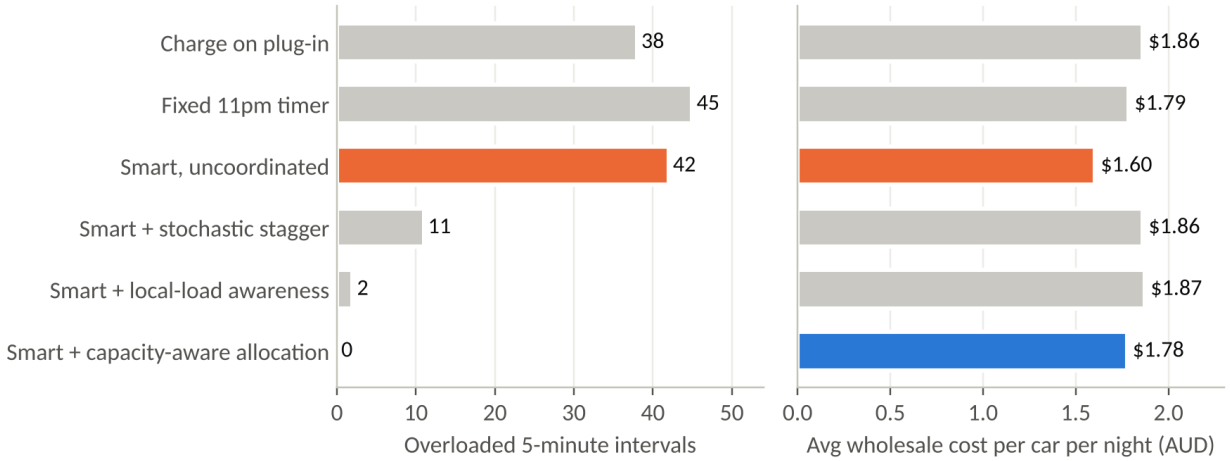


Figure 4. Overloaded intervals and average wholesale cost per car per night, by strategy.

What the results show

1. Individually optimal is collectively worst. Smart, uncoordinated charging produced the highest peak of all six strategies and a coincidence factor of exactly 1.00: every car charging in the same interval. Each car's choice is correct for that car, and it is the same choice every other car makes, because they all read the same prices. It is no safer for the network than a fixed 11pm timer; it only moves the spike.

2. Staggering on a statewide signal helps but is not enough. Using the Grid Health Score as a probability dial broke the perfect synchronisation and cut the peak to 114%, but 11 overloaded intervals remained. The score is built from statewide price and demand, so it says nothing about one street's transformer: two neighbours can both pick a "healthy" interval and stack on the same box.

3. Only local-capacity awareness clears it. Decentralised awareness of the local transformer took the peak to 100.8% with 2 residual overloads, both from cars forced on by their deadlines. The capacity-aware central allocator guaranteed zero overloads at 99.9% of rating. Overloads fell from 42 to 11 to 2 to 0 as each layer was added, and every strategy met 100% of deadlines.

4. Coordination is cheap. The capacity-aware strategy cost $1.78 per car per night against $1.60 for selfish smart charging: about 17 cents more, and still cheaper than charging on plug-in.

Supporting scenarios

Home batteries. A common objection is that home batteries make the problem go away. With 50% of homes carrying a 10 kWh battery, the overnight EV peak was unchanged. A battery shaves its own household's load; it does not communicate with the house next door. The EV (around 60 kWh) is much larger than the home battery (around 10 kWh) and still charges from the grid through the shared transformer.

Signal versus local load. A dedicated comparison (phase3_out/signal_vs_local.png) shows the statewide signal looking healthy at moments when the local transformer is already overloaded, which is the mechanism behind finding 2.

9. The cost question

The project was first pitched on driver savings. Measuring the numbers changed that framing.

- On a captured overnight window, smart timing averaged about $58/MWh against $66/MWh for charge-on-plug-in: roughly 12% cheaper on the wholesale component.
- In the simulation, uncoordinated smart charging cost $1.60 per car per night against $1.86 for charging on plug-in: about 26 cents.
- On a flat Queensland retail tariff, the most common kind, smart timing saves nothing at all, because the rate is the same at every hour. The saving exists only on wholesale-passthrough or time-of-use plans.

> If the individual saving is that small, the value has to be somewhere else. It is in the network: what happens when a whole suburb charges at once. That is a measurable contribution a single-household savings app cannot claim, and it is the reason the project is framed around coordination.

10. Phase 4: Hardware demonstrator

Hardware

Item	Detail
Microcontroller	Freenove ESP32-S3 (8 MB PSRAM), programmed in Arduino C++
Firmware libraries	WiFi, HTTPClient, ArduinoJson
Outputs	One LED on GPIO 2 for the live demo; ten blue LEDs on GPIO 1, 2, 4, 5, 6, 7, 15, 16, 17 and 18 for the ten-house demo; onboard RGB LED (GPIO 48) as the transformer indicator
Each LED circuit	GPIO to 220 ohm resistor to LED to shared ground rail
Input	Onboard BOOT button (GPIO 0) switches strategy
Server	Flask on the development laptop, port 5001, same Wi-Fi network

Stage 1: live decision

The ESP32 polls the server's /decision endpoint over Wi-Fi, parses the JSON with ArduinoJson and drives the LED from the charge field. The server calls AEMO, runs decision.py and returns price, demand, Grid Health Score, action and reason. AEMO updates every 5 minutes while the board polls every few seconds, so a 120-second cache sits between them: one upstream call per settlement interval instead of dozens of redundant ones. If AEMO is unreachable, the server serves the last good payload rather than failing.

The chain was validated end to end against AEMO's own dashboard. On 31 July 2026 at 13:45, AEMO showed a Queensland spot price of $48.36/MWh and demand of 6,188 MW; the server reported the same values, a CHARGE decision and a Grid Health Score of 7, and the LED was lit.

Stage 2: ten-house demonstrator

Ten LEDs represent ten houses on one transformer that can carry six chargers at once. suburb_demo.py replays the real night of 2 July 2026 from 9pm to the 7am deadline, compressed to about a minute. Each house plugs in between 9pm and 10:30pm and needs 18 to 32 kWh from a 7 kW charger. The onboard RGB LED is the transformer: green within capacity, red when overloaded. Pressing BOOT switches strategy and restarts the night, so the two runs are directly comparable.

Strategy	Peak cars at once	Overloaded intervals	Deadlines met	Cost / car
Smart, uncoordinated	10 of 10	42	10 / 10	$1.36
Coordinated (capacity-aware)	6	0	10 / 10	$1.52

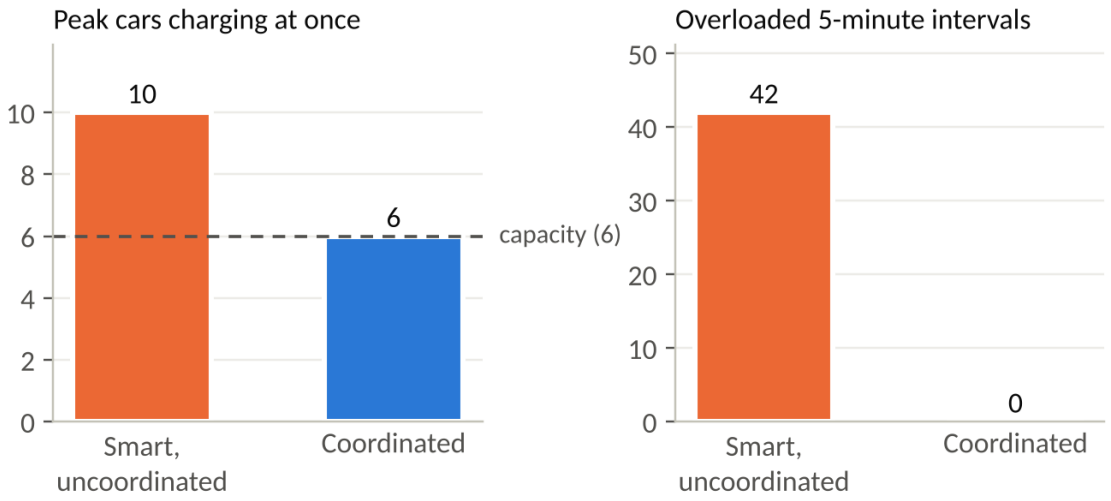


Figure 5. Ten-house demonstrator on the real night of 2 July 2026. The uncoordinated strategy exceeds capacity for 42 intervals; coordination never does.

The scaled-down demonstrator reproduces the shape of the Phase 3 result: uncoordinated smart charging puts every car on at once, while coordination keeps the transformer inside its rating, every car still finishes, and the cost difference is small. A video of the demonstrator is available on request.

11. Engineering lessons

Several problems during the build were worth recording because each looked like a code bug and was not.

Problem	Cause	Resolution
Serial monitor showed only garbage	The ESP32-S3 has two USB paths. With USB CDC On Boot enabled, output goes to the native USB port, not the UART bridge the cable was in.	Use the native USB port with CDC enabled, at 115200 baud.
Wi-Fi join looped forever	Plain WiFi.begin() gives no diagnostic. A WiFi.onEvent handler exposed disconnect reason 15, an authentication failure.	Corrected the password. The event handler is now the first debugging step.
Board could not see the network	The ESP32 radio is 2.4 GHz only.	Use the router's 2.4 GHz band.
Port 5000 unavailable	macOS AirPlay Receiver occupies port 5000.	Server runs on port 5001.
Port 5001 "already in use"	Closing a terminal window left the Flask process running.	Stop with Ctrl+C; clear strays with lsof.
New LEDs stayed dark	LEDs inserted backwards.	Long leg toward the GPIO side. Verified with a sweep test across all ten.
Demo state inconsistent	Flask's debug reloader runs two processes, each with its own demo clock.	Reloader disabled for the demo server.
Credentials at risk of publication	Wi-Fi password in sketch source.	Moved to a gitignored secrets.h with a committed template; repository history checked before going public.

12. Limitations

- **Known-price planner.** The per-car planner assumes it knows future prices. That is valid when replaying historical data and gives a best-case benchmark, but a live system would need AEMO forecasts plus an urgency override as a deadline approaches.
- **One dataset.** The simulation uses about 38 hours of Queensland winter data. Results should be checked across seasons, weekdays and extreme events before being generalised.
- **Simplified network model.** The transformer is modelled as a single kW limit. Voltage, phase imbalance and thermal time constants are not modelled.
- **Uniform fleet.** Charger power, energy needs and plug-in behaviour are simplified compared with real households.
- **Not connected to real chargers.** Deployment would need OCPP integration, CSIP-AUS compliance and network approval.
- **The demonstrator is an illustration.** Ten houses on a breadboard are a scaled visual of the simulation, not evidence in their own right.

13. Future work

- **Deployed-practice baseline.** Add a strategy that models how a dynamic operating envelope throttles chargers today (a uniform cap with no deadline awareness), so the comparison is against what networks actually deploy rather than against doing nothing.
- **Fairness metrics.** Measure who waits longest and whether the same households always lose out, addressing the question RACE for 2030 has flagged.
- **EV penetration sweep.** Find the adoption level at which each strategy first overloads the transformer, turning the result into a hosting-capacity estimate.
- **Forecast-driven planner.** Replace the known-price planner with AEMO forecasts plus an urgency override.
- **Multi-season data.** Extend from one capture to several weeks of historical AEMO data.
- **Decision engine tuning.** Recalibrate the demand band to observed Queensland demand and add hysteresis (for example start below $45/MWh, stop above $55/MWh) to prevent flicker near the threshold, which was observed live at $48.36/MWh.

14. Conclusion

Solar Spill set out to schedule EV charging into cheap Queensland hours and found that doing so independently, for every car, is the worst outcome for the shared network. On real Queensland data, uncoordinated smart charging pushed a 500 kVA transformer to 157% of its rating with 42 overloaded intervals. Coordination aware of the transformer's capacity held it to 99.9% with no overloads, every car still charged on time, at a cost of about 17 cents per car per night.

The broader lesson is that the problem is not timing but coordination. Price signals are already commoditised and individual savings are small; the hard, valuable problem is sharing scarce local capacity across many households. The project demonstrates that problem end to end: from live AEMO data, through a decision engine and a 100-home simulation, to a physical demonstrator that shows the result on a desk.

Appendix A. Repository and how to run

All code, data outputs and documentation are at **github.com/SSV227/solar-spill**.

```
git clone https://github.com/SSV227/solar-spill.git
cd solar-spill
python3 -m venv venv && source venv/bin/activate
pip install -r requirements.txt

python price_check.py     # one live AEMO reading
python decision.py        # decision engine self-test
python simulation.py      # full simulation; plots to phase3_out/
python suburb_demo.py     # ten-house results, smart vs coordinated
python server.py          # Flask server for the ESP32, port 5001
```

The ESP32 sketches need a secrets.h with Wi-Fi details (template: secrets.h.example), the server address set to the laptop's IP, and the ArduinoJson library. Board settings and wiring are in HARDWARE.md; threshold reasoning is in DECISIONS.md; simulation method is in PHASE3.md.

Appendix B. References

[1] AEMO, ELEC_NEM_SUMMARY public data feed. visualisations.aemo.com.au/aemo/apps/api/report/ELEC_NEM_SUMMARY

[2] Energex and Ergon Energy Network, *EV SmartCharge Queensland* trial findings.

[3] Energex, *EV Charging and QECM V4:2024*. energex.com.au/__data/assets/pdf_file/0003/1171884/EV-Charging-and-QECM-V4-2024.pdf

[4] Energex, EV charging and connections. energex.com.au/contractors/electrical-contractors/ev-charging-and-connections

[5] SolarQuotes, *Amber Pulls The Plug On Charge HQ*, July 2026. solarquotes.com.au/blog/amber-pulls-plug-charge-hq/

[6] Zecar, *Amber for EVs adds smart charging for MG, XPeng, Audi*, August 2026. zecar.com/reviews/amber-for-evs-mg-xpeng-audi-smart-charging

[7] EV Infrastructure News, *Australia's federal budget exposes low-voltage visibility gap as EV adoption surges*, May 2026.

[8] ESD News, *Managing low-voltage networks in Australia's distributed energy transition*, March 2026.

[9] RACE for 2030, *Low voltage network visibility and optimising DER hosting capacity*, 2021.